Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side**	*HTML5, CSS3, Jinja2 Templates*	*Provides a responsive and user-friendly interface for registration, login, dashboards, and loan prediction.*

2	**Backend / Server-Side**	*Python, Flask, Flask-SQLAlchemy*	*Handles business logic, user authentication, routing, database operations, and machine learning prediction.*
3	**Database / Data Storage**	*SQLite, CSV Dataset*	*SQLite stores user accounts, roles, and prediction history, while CSV files are used for training the machine learning model.*
4	**Cloud / Hosting / Deployment**	*Render*	Enables collaborative development, version control, branch management, and code review through pull requests.
5	**Version Control & CI/CD**	*Git, GitHub*	Enables collaborative development, version control, branch management, and code review through pull requests.

6	**Third-Party APIs / Other Tools**	*Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib, Werkzeug*	*Used for machine learning, data preprocessing, visualization, model persistence, and secure password hashing.*